

MAT21-4

Discrete Stochastic Models & Concentration

Conditional expectation, random walks, Markov chains, and concentration inequalities

From conditioning as information to quantitative bounds on uncertainty.

Albert School — 22 hours

Preface

You have already met probability twice. In **MAT11-4 (Discrete Mathematics)** you learned to reason over **partitions**: an equivalence relation cuts a set into disjoint classes, and a weighted average over those classes — “given case A , what fraction has property B ?” — is exactly the bookkeeping that the **law of total probability** later formalised. In **MAT21-2 (Probability & Statistics I)** you turned that bookkeeping into a theory: probability spaces, conditional probability, Bayes’ theorem, expectation and variance of discrete random variables, and the classic distributions (Bernoulli, binomial, geometric, Poisson). From **MAT11-3 (Linear Algebra I)** you carry matrix arithmetic and the idea that iterating a linear map means taking matrix powers — the engine of Markov chains.

This course deepens that foundation in three movements.

First, we sharpen **conditioning**. In MAT21-2 you conditioned on a single event; here we condition on an entire **partition** at once and obtain a new random variable, the **conditional expectation** $\mathbb{E}[X \mid \mathcal{P}]$. This single object organises the **tower property**, the **law of total variance**, and the “fair game” idea behind **martingales**.

Second, we study two concrete families of **stochastic models**: **random walks** (gambler’s ruin, hitting times, return probabilities) and **Markov chains** (transition matrices, recurrence and transience, stationary distributions). These are the discrete prototypes of every dynamic random process you will meet later.

Third, we learn to **bound uncertainty quantitatively**. Markov’s, Chebyshev’s, and Hoeffding’s inequalities convert a fact about a mean or variance into a guarantee about how rare large deviations are — and, read backwards, into a **sample-size requirement**: how many observations buy a given confidence at a given tolerance.

A word on rigour and scope. We work entirely on **finite or countable** probability spaces, so every expectation is a finite (or convergent) sum and every manipulation is elementary. We deliberately **avoid** the formal machinery of sigma-algebras, filtrations, and measure-theoretic martingales: that apparatus is built from scratch in S6. Here, a “partition” carries all the information we need, and a martingale is simply a fair game. Throughout, each idea appears first on a small concrete example, then as a general statement; **common pitfalls** are flagged in red boxes; and **Python mini-labs** let you watch the theory hold up against simulation.

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1 Conditional Expectation on Finite Partitions

1.1 From conditioning on an event to conditioning on a partition

In MAT21-2 you computed $\mathbb{E}[X \mid A]$, the expectation of X given that a **single** event A occurred. The new idea of this course is to condition on a whole **partition** of the sample space at once, and to package the answer not as a number but as a **random variable**.

Worked example — a single die. Roll a fair die, $\Omega = \{1, 2, 3, 4, 5, 6\}$, each outcome with probability $1/6$. Let X be the number shown. Partition Ω by **parity**:

$$\mathcal{P} = \{A_{\text{odd}}, A_{\text{even}}\}, \quad A_{\text{odd}} = \{1, 3, 5\}, \quad A_{\text{even}} = \{2, 4, 6\}.$$

Conditioning on each block separately,

$$\mathbb{E}[X \mid A_{\text{odd}}] = \frac{1 + 3 + 5}{3} = 3, \quad \mathbb{E}[X \mid A_{\text{even}}] = \frac{2 + 4 + 6}{3} = 4.$$

Now define a new random variable Y that reports, for each outcome ω , the average over the block containing ω :

$$Y(\omega) = \begin{cases} 3 & \text{if } \omega \text{ is odd} \\ 4 & \text{if } \omega \text{ is even} \end{cases}.$$

This Y is the **conditional expectation of X given the partition $\mathcal{P}$** . It is a coarser, “smoothed” version of X : it forgets the exact value and remembers only the block average.

The defining feature is that Y is **constant on each block** — it depends only on **which** block you are in, never on where inside the block. That is what “knowing only the information in $\mathcal{P}$ ” means.

Definition 1 (conditional expectation given a partition) Let $(\Omega, \mathbb{P})$ be a finite probability space, $X : \Omega \rightarrow \mathbb{R}$ a random variable, and $\mathcal{P} = \{B_1, \dots, B_k\}$ a partition of Ω into events of positive probability. The **conditional expectation of X given $\mathcal{P}$** is the random variable $\mathbb{E}[X \mid \mathcal{P}]$ defined by

$$\mathbb{E}[X \mid \mathcal{P}](\omega) = \mathbb{E}[X \mid B_j] = \frac{\sum_{\omega' \in B_j} X(\omega') \mathbb{P}(\{\omega'\})}{\mathbb{P}(B_j)} \quad \text{for every } \omega \in B_j.$$

It is constant on each block B_j , taking there the value $\mathbb{E}[X \mid B_j]$.

Remark We write $\mathbb{E}[X \mid Z]$ as shorthand for $\mathbb{E}[X \mid \mathcal{P}_Z]$, where $\mathcal{P}_Z$ is the partition of Ω into the level sets $\{Z = z\}$ of a discrete random variable Z . “Conditioning on Z ” means “conditioning on the partition by the value of Z .” On the block $\{Z = z\}$ the conditional expectation equals the ordinary number $\mathbb{E}[X \mid Z = z]$.

1.2 The two faces: a number and a random variable

Keep two objects strictly apart, because confusing them is the single most common error in this chapter.

- $\mathbb{E}[X \mid Z = z]$ is a **number**: the average of X over the block where $Z = z$.
- $\mathbb{E}[X \mid Z]$ is a **random variable**: the function whose value at ω is $\mathbb{E}[X \mid Z = Z(\omega)]$.

Common pitfall $\mathbb{E}[X | Z]$ is **not a number** and **not a function of X alone**. It is a random variable that is constant on the blocks $\{Z = z\}$. Writing “ $\mathbb{E}[X | Z] = 3.5$ ” for the die above is wrong: the correct answer is the **two-valued** variable Y equal to 3 on odds and 4 on evens. A plain number only appears once you fix a value z (then $\mathbb{E}[X | Z = z]$) or once you take a **further** expectation (then $\mathbb{E}[\mathbb{E}[X | Z]] = \mathbb{E}[X]$, see below).

	odd = {1, 3, 5}			even = {2, 4, 6}		
X :	1	2	3	4	5	6
$\mathbb{E}[X \mathcal{P}]$:	3	4	3	4	3	4

Figure 1: Conditioning by a partition replaces X (top, six distinct values) with its block averages $\mathbb{E}[X | \mathcal{P}]$ (bottom, constant on each block).

1.3 Computation by partition refinement

The block-average definition is also a **recipe**. Conditioning is most powerful when you condition step by step, refining a coarse partition into a finer one.

Worked example — two stages. A bag has 2 fair coins and 1 biased coin with $\mathbb{P}(\text{heads}) = 0.8$. You pick a coin uniformly at random, then flip it. Let $X = 1$ if heads, else 0. Partition first by which coin was drawn: $\mathcal{P} = \{\text{fair}_1, \text{fair}_2, \text{biased}\}$, each of probability $1/3$. On each block,

$$\mathbb{E}[X | \text{fair}] = 0.5, \quad \mathbb{E}[X | \text{biased}] = 0.8.$$

So $\mathbb{E}[X | \mathcal{P}]$ equals 0.5 on the two fair-coin blocks and 0.8 on the biased block. Averaging these block values weighted by block probability recovers the unconditional mean — the next section makes this exact.

Definition 2 (refinement) A partition $\mathcal{Q}$ **refines** a partition $\mathcal{P}$ (written $\mathcal{Q} \geq \mathcal{P}$) if every block of $\mathcal{Q}$ is contained in some block of $\mathcal{P}$ — i.e. $\mathcal{Q}$ is obtained by **subdividing** the blocks of $\mathcal{P}$. A finer partition carries **more information**.

In the coin example, “which coin **and** what it landed” is a refinement of “which coin.” Refining never destroys information; it can only split blocks further.

1.4 The tower property

What happens if we condition on a coarse partition the conditional expectation already computed for a fine one? We get the coarse answer directly — the finer detail washes out. This is the **tower property** (also called the **law of iterated expectations**).

Theorem 3 (tower property) Let $\mathcal{P}$ and $\mathcal{Q}$ be partitions with $\mathcal{Q} \geq \mathcal{P}$ (so $\mathcal{Q}$ is finer). Then for any random variable X ,

$$\mathbb{E}[\mathbb{E}[X | \mathcal{Q}] | \mathcal{P}] = \mathbb{E}[X | \mathcal{P}].$$

In particular, taking $\mathcal{P} = \{\Omega\}$ the trivial partition (which gives the plain mean),

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{Q}]] = \mathbb{E}[X].$$

Proof. Fix a block $B \in \mathcal{P}$. Because $\mathcal{Q}$ refines $\mathcal{P}$, B is a disjoint union of blocks $C_1, \dots, C_m \in \mathcal{Q}$. On each C_i the inner variable $\mathbb{E}[X \mid \mathcal{Q}]$ is constant, equal to $\mathbb{E}[X \mid C_i]$. Averaging that over B ,

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{Q}] \mid B] = \sum_{i=1}^m \mathbb{P}(C_i \mid B) \cdot \mathbb{E}[X \mid C_i] = \sum_{i=1}^m \frac{\mathbb{P}(C_i)}{\mathbb{P}(B)} \cdot \frac{\sum_{\omega \in C_i} X(\omega) \mathbb{P}(\{\omega\})}{\mathbb{P}(C_i)}.$$

The $\mathbb{P}(C_i)$ cancel, and the double sum collects every $\omega \in B$ exactly once:

$$= \frac{1}{\mathbb{P}(B)} \sum_{\omega \in B} X(\omega) \mathbb{P}(\{\omega\}) = \mathbb{E}[X \mid B].$$

Since this holds on every block $B \in \mathcal{P}$, the two random variables agree everywhere. □

Remark Read the tower property as **consistency of averaging**: you may compute a block average all at once, or first average over a finer subdivision and then average those sub-averages weighted by their sizes — the result is identical. The order “refine then coarsen” gives the coarse answer, never something in between.

Common pitfall The tower property requires **nesting**: one partition must refine the other. For two **unrelated** partitions $\mathcal{P}, \mathcal{Q}$ (neither refining the other), $\mathbb{E}[\mathbb{E}[X \mid \mathcal{Q}] \mid \mathcal{P}]$ generally differs from both $\mathbb{E}[X \mid \mathcal{P}]$ and $\mathbb{E}[X \mid \mathcal{Q}]$. Always check the nesting before collapsing a double conditional.

1.5 Conditioning as information gain

Conditioning on a finer partition is **learning more**. Two limiting cases pin down the scale:

- The **trivial partition** $\mathcal{P} = \{\Omega\}$ carries no information: $\mathbb{E}[X \mid \{\Omega\}] = \mathbb{E}[X]$, a constant.
- The **finest partition** into singletons $\{\{\omega\} : \omega \in \Omega\}$ carries full information: $\mathbb{E}[X \mid \text{singletons}] = X$ itself.

Every partition sits between these extremes. The finer it is, the closer $\mathbb{E}[X \mid \mathcal{P}]$ tracks X . In Chapter 2 we make “closer” quantitative: the finer the partition, the more of X ’s variance the conditional expectation explains.

Worked example — conditioning can only help on average. Let X be a fair die and $\mathcal{P}$ the parity partition. Then $\mathbb{E}[X] = 3.5$, while $\mathbb{E}[X \mid \mathcal{P}]$ is 3 or 4 — already a better guess of X once you are told the parity. Knowing the block shrinks your uncertainty about X ; that is the sense in which conditioning is **information gain**.

1.6 Mini-lab: verifying the tower property in Python

```
import numpy as np
rng = np.random.default_rng(0)
```

```
# X = sum of two dice; Z = value of the first die (the coarser information)
N = 2_000_000
d1 = rng.integers(1, 7, N)
d2 = rng.integers(1, 7, N)
```

```

X = d1 + d2
Z = d1

# E[X | Z=z] for each z, then the random variable E[X|Z](omega)
cond = {z: X[Z == z].mean() for z in range(1, 7)}
EX_given_Z = np.array([cond[z] for z in Z])

print("E[X]                =", X.mean())           # ~ 7.0
print("E[ E[X|Z] ]         =", EX_given_Z.mean())   # ~ 7.0 (tower)
print("E[X|Z=1], E[X|Z=6]  =", cond[1], cond[6])    # ~ 4.5, ~ 9.5

```

The averaged conditional expectation reproduces $\mathbb{E}[X] = 7$ to three digits, while the block values $\mathbb{E}[X | Z = z] = z + 3.5$ confirm the exact formula. This is the tower property, observed.

Exercises

1. Roll a fair die and let X be its value. For the partition $\mathcal{P} = \{\{1, 2\}, \{3, 4\}, \{5, 6\}\}$, write $\mathbb{E}[X | \mathcal{P}]$ explicitly as a three-valued random variable, and verify $\mathbb{E}[\mathbb{E}[X | \mathcal{P}]] = 3.5$.
2. Draw a card from a standard 52-card deck. Let X be its value (Ace= 1, ..., King= 13) and condition on **suit**. What is $\mathbb{E}[X | \text{suit}]$? Is it constant? Explain why in terms of symmetry.
3. Let $\mathcal{Q} \geq \mathcal{P}$. Prove that $\mathbb{E}[X | \mathcal{P}]$ and $\mathbb{E}[\mathbb{E}[X | \mathcal{P}] | \mathcal{Q}]$ are equal. (This is the “other order”: coarsen then refine. Why does conditioning a partition-measurable variable on a **finer** partition leave it unchanged?)
4. Give two partitions $\mathcal{P}, \mathcal{Q}$ of a four-point space and a variable X such that $\mathbb{E}[\mathbb{E}[X | \mathcal{Q}] | \mathcal{P}] \neq \mathbb{E}[\mathbb{E}[X | \mathcal{P}] | \mathcal{Q}]$, demonstrating that without nesting the tower property fails.
5. Two fair coins are tossed. Let X be the number of heads and $Z = 1$ if the first toss is heads, else 0. Compute $\mathbb{E}[X | Z]$ as a random variable and check the tower property numerically.

2 Variance Decomposition

2.1 Two ways to average: the law of total expectation

The tower property, written for the trivial outer partition, is so useful it earns its own name.

Theorem 4 (law of total expectation) For any random variable X and any partition $\mathcal{P} = \{B_1, \dots, B_k\}$,

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X | \mathcal{P}]] = \sum_{j=1}^k \mathbb{P}(B_j) \mathbb{E}[X | B_j].$$

This is the partition-and-condition reasoning of MAT11-4 made exact: chop the sample space into informative cases, average within each, then average the case averages weighted by their probabilities.

Worked example — expected claim. An insurer’s customers are 70% low-risk (expected claim €200) and 30% high-risk (expected claim €900). The expected claim of a random customer is

$$\mathbb{E}[X] = 0.7 \times 200 + 0.3 \times 900 = 140 + 270 = 410 \text{ €}.$$

The partition into risk classes makes a hard average easy.

2.2 The law of total variance

Expectation splits cleanly across a partition. Variance does **not** — and the correction term is the whole point. Conditioning explains **part** of the variance; what remains is the leftover scatter inside the blocks.

Theorem 5 (law of total variance) For any random variable X with finite variance and any partition $\mathcal{P}$,

$$\underbrace{\text{Var}(X)}_{\text{total}} = \underbrace{\mathbb{E}[\text{Var}(X | \mathcal{P})]}_{\text{within-block (residual)}} + \underbrace{\text{Var}(\mathbb{E}[X | \mathcal{P}])}_{\text{between-block (explained)}}.$$

Here $\text{Var}(X | \mathcal{P})$ is the random variable equal to the ordinary variance of X computed inside each block.

Proof. Write $m = \mathbb{E}[X | \mathcal{P}]$ and use $\text{Var}(X | \mathcal{P}) = \mathbb{E}[X^2 | \mathcal{P}] - m^2$. Take expectations and apply the tower property ($\mathbb{E}[\mathbb{E}[X^2 | \mathcal{P}]] = \mathbb{E}[X^2]$, $\mathbb{E}[m] = \mathbb{E}[X]$):

$$\mathbb{E}[\text{Var}(X | \mathcal{P})] = \mathbb{E}[X^2] - \mathbb{E}[m^2].$$

Also $\text{Var}(m) = \mathbb{E}[m^2] - (\mathbb{E}[m])^2 = \mathbb{E}[m^2] - (\mathbb{E}[X])^2$. Adding,

$$\mathbb{E}[\text{Var}(X | \mathcal{P})] + \text{Var}(m) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \text{Var}(X). \square$$

□

Remark The decomposition is the probabilistic ANOVA you may meet in statistics: **total = within + between**, or “**unexplained + explained**.” The term $\text{Var}(\mathbb{E}[X | \mathcal{P}])$ is the variance the partition **explains** — how much X moves as you change blocks; $\mathbb{E}[\text{Var}(X | \mathcal{P})]$ is the variance it leaves **unexplained** — the average scatter still present inside a block.

Worked example — mixed exam scores. Class A (60% of students) scores with mean 12 and variance 4; class B (40%) with mean 15 and variance 9. Let $\mathcal{P}$ be the class partition.

Overall mean: $\mathbb{E}[X] = 0.6 \times 12 + 0.4 \times 15 = 13.2$.

Within-block (residual): $\mathbb{E}[\text{Var}(X | \mathcal{P})] = 0.6 \times 4 + 0.4 \times 9 = 6.0$.

Between-block (explained): the variable $\mathbb{E}[X | \mathcal{P}]$ takes value 12 with prob 0.6 and 15 with prob 0.4, so its variance is

$$0.6(12 - 13.2)^2 + 0.4(15 - 13.2)^2 = 0.6(1.44) + 0.4(3.24) = 0.864 + 1.296 = 2.16.$$

Total variance: $\text{Var}(X) = 6.0 + 2.16 = 8.16$. The class label explains $2.16/8.16 \approx 26\%$ of the total variation; the remaining 74% is scatter within classes.

Common pitfall Variance does **not** satisfy a naive “law of total variance” without the explained term. Writing $\text{Var}(X) = \mathbb{E}[\text{Var}(X | \mathcal{P})]$ (forgetting $\text{Var}(\mathbb{E}[X | \mathcal{P}])$) **understates** the true variance whenever the block means differ. Both non-negative terms are always present; dropping either is a sign error waiting to happen.

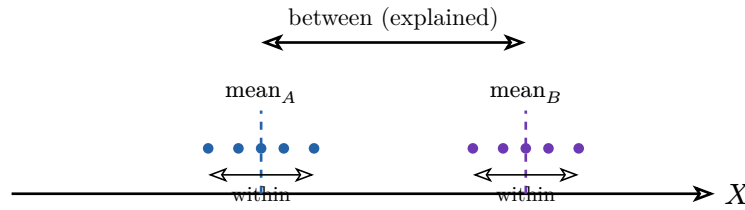


Figure 2: Total variation splits into **between-block** spread of the block means (explained) and **within-block** scatter (residual). Wider separation of means raises the explained share.

2.3 Refinement increases the explained variance

Because the residual term $\mathbb{E}[\text{Var}(X \mid \mathcal{P})]$ is non-negative and the total is fixed, **finer partitions explain more**. Conditioning on more information can only reduce (never increase) the average residual variance.

Proposition 6 (conditioning reduces residual variance) If $\mathcal{Q} \geq \mathcal{P}$ (i.e. $\mathcal{Q}$ is finer), then

$$\mathbb{E}[\text{Var}(X \mid \mathcal{Q})] \leq \mathbb{E}[\text{Var}(X \mid \mathcal{P})],$$

with equality iff $\mathbb{E}[X \mid \mathcal{Q}] = \mathbb{E}[X \mid \mathcal{P}]$. Equivalently, the explained variance $\text{Var}(\mathbb{E}[X \mid \mathcal{P}])$ is non-decreasing under refinement.

This is the precise sense in which “more information = less leftover uncertainty.” At the extreme, conditioning on singletons leaves zero residual: $\mathbb{E}[X \mid \text{singletons}] = X$, so $\text{Var}(\mathbb{E}[X \mid \text{singletons}]) = \text{Var}(X)$ — the partition explains everything.

2.4 Mini-lab: the variance decomposition, checked

```
import numpy as np
rng = np.random.default_rng(1)

# Mixture: class A ~ Normal(12, 2^2), class B ~ Normal(15, 3^2)
N = 2_000_000
is_A = rng.random(N) < 0.6
X = np.where(is_A, rng.normal(12, 2, N), rng.normal(15, 3, N))

total = X.var()
mA, mB = X[is_A].mean(), X[~is_A].mean()
within = is_A.mean()*X[is_A].var() + (~is_A).mean()*X[~is_A].var()
cond_mean = np.where(is_A, mA, mB)
between = cond_mean.var()

print(f"total    = {total:.3f}")           # ~ 8.16
print(f"within   = {within:.3f}")           # ~ 6.0   (residual)
print(f"between  = {between:.3f}")           # ~ 2.16  (explained)
print(f"sum      = {within + between:.3f}")
```

The simulated `within + between` matches `total` to two decimals, reproducing the worked example.

Exercises

6. A factory runs two lines. Line 1 (75% of output) has mean length 50.0 mm, sd 0.2 mm; line 2 (25%) has mean 50.6 mm, sd 0.3 mm. Compute the overall mean and the total variance via the decomposition, and report the fraction of variance explained by “which line.”

7. Let N be $\text{Poisson}(\lambda)$ and, given $N = n$, let X be $\text{Binomial}(n, p)$ (e.g. N eggs laid, each hatching with probability p). Use the laws of total expectation and total variance to show $\mathbb{E}[X] = \lambda p$ and $\text{Var}(X) = \lambda p$. (Recall $\mathbb{E}[N] = \text{Var}(N) = \lambda$.)
8. Show that the explained variance is zero iff all block means $\mathbb{E}[X \mid B_j]$ are equal, and that the residual variance is zero iff X is constant on each block.
9. Prove the proposition on refinement: with $\mathcal{Q} \geq \mathcal{P}$, apply the law of total variance **inside each block of $\mathcal{P}$** to the finer partition $\mathcal{Q}$, then average.
10. (Conditioning need not reduce variance pointwise.) Find X and a block B with $\text{Var}(X \mid B) > \text{Var}(X)$. Reconcile this with the proposition — which quantity is guaranteed to decrease, the conditional variance on a **single** block or its **average**?

3 Martingales: The Fair-Game Property

3.1 A fair game

A **martingale** is the mathematical model of a **fair game**: at every step, your expected fortune after the next play, given everything you know now, equals your fortune now. You neither expect to win nor to lose.

Worked example — doubling on red. You hold €100 and bet on repeated fair coin tosses, winning or losing your stake. Let S_n be your fortune after n tosses with a fixed €1 stake. Knowing the whole history up to time n , the next toss adds $+1$ or -1 with equal probability, so

$$\mathbb{E}[S_{n+1} \mid \text{history up to } n] = S_n + \frac{1}{2}(+1) + \frac{1}{2}(-1) = S_n.$$

On average you stand still: the game is fair. The sequence (S_n) is a martingale.

We capture “everything you know now” with a partition that gets finer over time, without invoking formal filtrations.

Definition 7 (martingale, partition form) Let $\mathcal{P}_0 \leq \mathcal{P}_1 \leq \mathcal{P}_2 \leq \dots$ be an increasing (refining) sequence of partitions of Ω — information accumulating over time — and let $(S_n)_{n \geq 0}$ be random variables with S_n constant on the blocks of $\mathcal{P}_n$ (i.e. S_n is known once you know the time- n information). The sequence is a **martingale** if $\mathbb{E}[|S_n|] < \infty$ for all n and

$$\mathbb{E}[S_{n+1} \mid \mathcal{P}_n] = S_n \quad \text{for every } n \geq 0.$$

If “=” is replaced by “ $\leq$ ” it is a **supermartingale** (unfavourable game); by “ $\geq$ ”, a **submartingale** (favourable game).

Remark Taking the unconditional expectation of the defining equation and using the tower property gives the headline consequence: a martingale has **constant mean**,

$$\mathbb{E}[S_n] = \mathbb{E}[S_0] \quad \text{for all } n.$$

A fair game neither gains nor loses **on average**, at any fixed time horizon.

3.2 Canonical examples

Worked example — symmetric random walk. Let $X_1, X_2, \dots$ be independent with $\mathbb{P}(X_i = +1) = \mathbb{P}(X_i = -1) = 1/2$, and $S_n = X_1 + \dots + X_n$ with $S_0 = 0$. Let $\mathcal{P}_n$ encode the first n steps. Then $\mathbb{E}[S_{n+1} \mid \mathcal{P}_n] = S_n + \mathbb{E}[X_{n+1}] = S_n$, so (S_n) is a martingale: the symmetric random walk is the prototypical fair game.

Worked example — a Doob martingale — revealing information gradually. Fix a random variable Y (a final payoff) and a refining sequence $\mathcal{P}_n$. Define $S_n = \mathbb{E}[Y \mid \mathcal{P}_n]$: your **best current estimate** of Y given what you have learned by time n . By the tower property (the partitions are nested),

$$\mathbb{E}[S_{n+1} \mid \mathcal{P}_n] = \mathbb{E}[\mathbb{E}[Y \mid \mathcal{P}_{n+1}] \mid \mathcal{P}_n] = \mathbb{E}[Y \mid \mathcal{P}_n] = S_n.$$

So **any** sequence of successively refined estimates of a fixed quantity is a martingale. Your running prediction of the final outcome is a fair game: today's estimate is the best forecast of tomorrow's estimate. This **Doob martingale** is the bridge to the concentration results of Chapter 6.

Worked example — a multiplicative martingale. For the **biased** walk with $\mathbb{P}(X_i = +1) = p$, $\mathbb{P}(X_i = -1) = q = 1 - p$, the sum S_n is **not** a martingale (its mean drifts). But $M_n = (q/p)^{S_n}$ **is**: since $\mathbb{E}[(q/p)^{X_{n+1}}] = p(q/p) + q(p/q) = q + p = 1$, independence gives $\mathbb{E}[M_{n+1} \mid \mathcal{P}_n] = M_n$. This exponential reweighting tames a biased walk into a fair game and will solve biased gambler's ruin in Chapter 4.

3.3 Optional stopping: the intuition

A **stopping time** τ is a random time at which you decide to quit using **only information available so far** — “stop the first time my fortune hits €200” is legal; “stop just before the last loss” is not (it peeks at the future).

The **optional stopping theorem** says that, under suitable conditions, **you cannot beat a fair game by clever quitting**: your expected fortune at the stopping time equals your starting fortune.

Theorem 8 (optional stopping, intuitive form) Let (S_n) be a martingale and τ a stopping time. If any one of the following holds:

1. τ is bounded ($\tau \leq N$ for a constant N), or
2. τ is finite with probability 1 and the increments are bounded ($|S_{n+1} - S_n| \leq c$), or
3. $\mathbb{E}[\tau] < \infty$ and the increments are bounded,

then

$$\mathbb{E}[S_\tau] = \mathbb{E}[S_0].$$

We do not prove this in its general measure-theoretic form (that is S6 material); the partition picture makes the **bounded** case believable: stopping by a fixed horizon N is just averaging the fair increments up to N , each of which contributes zero on average.

Remark Read optional stopping as a **no-free-lunch** theorem: any stopping rule built from past information leaves the expected payoff of a fair game unchanged. To gain, you

would need to foresee the future or play infinitely with unlimited stakes — exactly what the conditions forbid.

3.4 When optional stopping fails: a counterexample

Drop a condition and the conclusion collapses. This is the most instructive single example in the chapter.

Worked example — the martingale betting system. Symmetric random walk $S_0 = 0$. Define $\tau =$ “the first time $S_n = +1$.” Since the symmetric walk is recurrent (Chapter 4), $\tau < \infty$ with probability 1, and **by definition** $S_\tau = 1$. Hence

$$\mathbb{E}[S_\tau] = 1 \neq 0 = \mathbb{E}[S_0].$$

Optional stopping **fails**. Which hypothesis broke? Not finiteness — τ is finite a.s. The culprit is the **unbounded waiting / unbounded exposure**: $\mathbb{E}[\tau] = \infty$, and to guarantee the eventual €1 gain you must tolerate arbitrarily large interim losses (and, in the casino version, arbitrarily large stakes). The “sure-win” doubling system is exactly this τ ; it appears to beat a fair game only by assuming a bottomless bankroll.

Common pitfall “ $\tau < \infty$ with probability 1” is **not enough** for $\mathbb{E}[S_\tau] = \mathbb{E}[S_0]$. You must also control the exposure — via boundedness of τ , of the increments **together with** $\mathbb{E}[\tau] < \infty$, or a uniform-integrability condition. The doubling system is the standard trap: every individual game is fair, yet a naive “I’ll just stop when ahead” reasoning predicts a guaranteed profit. The flaw is the unbounded $\mathbb{E}[\tau]$ and unbounded losses along the way, not any single unfair bet.

3.5 Mini-lab: a fair game stays fair (until you cheat with unbounded stakes)

```
import numpy as np
rng = np.random.default_rng(2)

def walk_until_plus_one(cap):
    """Symmetric walk; stop at +1 OR when n hits cap (bounded stopping)."""
    s, n = 0, 0
    while s != 1 and n < cap:
        s += 1 if rng.random() < 0.5 else -1
        n += 1
    return s

for cap in (10, 100, 1000, 10_000):
    vals = [walk_until_plus_one(cap) for _ in range(20_000)]
    print(f"cap={cap:6d}:  E[S_tau] ~ {np.mean(vals): .3f}")
```

With a **bounded** horizon cap , optional stopping applies and $\mathbb{E}[S_\tau] \approx 0$. As cap grows, more runs reach +1 before the cutoff and the average creeps toward 1 — the unbounded- τ regime where optional stopping breaks. The “guaranteed €1” only materialises in the limit of unlimited patience and bankroll.

Exercises

11. Verify that the biased-walk product $M_n = (q/p)^{S_n}$ satisfies $\mathbb{E}[M_{n+1} \mid \mathcal{P}_n] = M_n$, and deduce $\mathbb{E}[M_n] = 1$ for all n .
12. Let (S_n) be the symmetric walk. Show $S_n^2 - n$ is a martingale. (Compute $\mathbb{E}[S_{n+1}^2 \mid \mathcal{P}_n]$ using $S_{n+1} = S_n \pm 1$.)
13. A gambler plays a game where each round multiplies the stake by 2 (prob 1/2) or 0 (prob 1/2), reinvesting everything. Is fortune a martingale? Compute $\mathbb{E}[S_n]$ and $\mathbb{P}(S_n > 0)$, and comment on the gap between “expected fortune” and “typical fortune.”
14. For the counterexample’s τ (“first hit +1”), explain in one paragraph which precise hypothesis of optional stopping fails, and why finiteness of τ alone is insufficient.
15. Give a **bounded** stopping time $\tau \leq 3$ for the symmetric walk and verify $\mathbb{E}[S_\tau] = 0$ directly by enumerating the 2^3 equally likely paths.

4 Random Walks

4.1 The model

A one-dimensional random walk starts at $S_0 = a$ and adds independent steps $X_i \in \{+1, -1\}$ with $\mathbb{P}(X_i = +1) = p$ and $\mathbb{P}(X_i = -1) = q = 1 - p$. When $p = 1/2$ the walk is **symmetric** (unbiased); otherwise it is **biased**, drifting right if $p > 1/2$ and left if $p < 1/2$. Random walks model a gambler’s bankroll, a queue length, a diffusing particle, or a stock’s tick-by-tick moves.

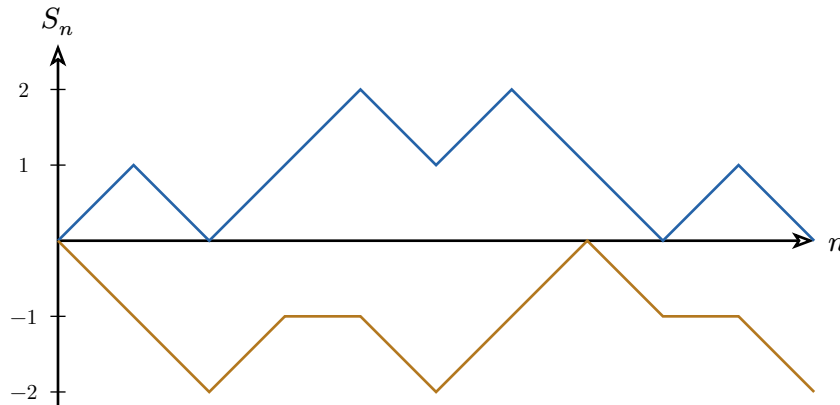


Figure 3: Two sample paths of a symmetric walk from $S_0 = 0$. The walk wanders with no drift; typical displacement after n steps grows like $\sqrt{n}$.

4.2 Gambler’s ruin

A gambler starts with $\text{€}a$ and bets $\text{€}1$ per round against a house, quitting on ruin (0) or on reaching a target $\text{€}N$. What is the probability of ruin? This is a random walk **absorbed** at 0 and N .

Theorem 9 (gambler’s ruin) Let $h(a) = \mathbb{P}(\text{reach } N \text{ before } 0 \mid S_0 = a)$ for $0 \leq a \leq N$. Then

$$h(a) = \begin{cases} \frac{a}{N} & \text{if } p = \frac{1}{2} \\ \frac{(q/p)^a - 1}{(q/p)^N - 1} & \text{if } p \neq \frac{1}{2}. \end{cases}$$

Proof. Condition on the first step (law of total expectation, one step of refinement):

$$h(a) = p h(a+1) + q h(a-1), \quad h(0) = 0, h(N) = 1.$$

This is a linear recurrence — exactly the closed-form-from-recurrence technique of MAT11-4. For $p = 1/2$ it reads $h(a) = (h(a+1) + h(a-1))/2$, i.e. h has constant slope (it is **harmonic**): the only solution with $h(0) = 0, h(N) = 1$ is the straight line $h(a) = a/N$. For $p \neq 1/2$ the characteristic equation $pr^2 - r + q = 0$ has roots $r = 1$ and $r = q/p$, so $h(a) = A + B(q/p)^a$; the boundary conditions fix A, B and give the stated ratio. $\square$

Worked example — a fair but doomed game. With $a = 50, N = 100, p = 1/2$: ruin probability $1 - h(a) = 1 - 50/100 = 1/2$. Fair. But raise the target to $N = 1000$ with the same start $a = 50$: now $h = 50/1000 = 0.05$, so ruin probability is 95%. Even in a **fair** game, chasing a distant target almost surely bankrupts you first — the asymmetry between your small stake and the large goal does the damage.

Worked example — the house edge compounds. European roulette red/black: $p = 18/37 \approx 0.486$, so $q/p \approx 1.0556$. Starting with €100 aiming to double to €200 ($a = 100, N = 200$):

$$h = \frac{1.0556^{100} - 1}{1.0556^{200} - 1} \approx \frac{1.0556^{100} - 1}{1.0556^{100} - 1} (1.0556^{100} + 1) \approx \frac{1}{1.0556^{100}} \approx 0.0044.$$

A mere 1.4% per-bet disadvantage collapses the chance of doubling to under 0.5%. Small biases compound brutally over many rounds.

Common pitfall A **fair** game ($p = 1/2$) does **not** mean a safe one. Ruin probability against a much richer opponent is near 1 because $h(a) = a/N$ is tiny when $N \gg a$. The relevant quantity is your bankroll **relative to the target / the opponent**, not the fairness of a single bet.

4.3 Expected duration

How long until the game ends? Let $D(a)$ be the expected number of steps starting from a .

Proposition 10 (expected ruin time) With absorption at 0 and N ,

$$D(a) = \begin{cases} a(N-a) & \text{if } p = \frac{1}{2} \\ \frac{a}{q-p} - \frac{N}{q-p} \cdot \frac{(q/p)^a - 1}{(q/p)^N - 1} & \text{if } p \neq \frac{1}{2}. \end{cases}$$

The symmetric formula $D(a) = a(N-a)$ peaks at the midpoint: from $a = 50, N = 100$ the fair game lasts 2500 steps on average — surprisingly long, because the walk drifts nowhere and meanders.

Proof. (symmetric case sketch) $\square$

D satisfies $D(a) = 1 + (D(a+1) + D(a-1))/2$ with $D(0) = D(N) = 0$. The inhomogeneous term forces a quadratic particular solution; $-a^2$ works, and adding the linear homogeneous solution to meet the boundary conditions yields $D(a) = a(N-a)$. $\square$

4.4 Hitting times and return probabilities on the unbounded walk

Remove the upper wall ($N \rightarrow \infty$) and ask about the walk on all of $\mathbb{Z}$.

Proposition 11 (recurrence of the symmetric walk) The symmetric ($p = 1/2$) walk on $\mathbb{Z}$ is **recurrent**: starting from 0 it returns to 0 with probability 1. The biased ($p \neq 1/2$) walk is **transient**: it returns with probability $1 - |p - q| < 1$, and otherwise drifts to $\pm\infty$.

Worked example — certain return, infinite wait. For the symmetric walk, let τ_0 be the time of first return to the origin. Then $\mathbb{P}(\tau_0 < \infty) = 1$ — return is certain — yet $\mathbb{E}[\tau_0] = \infty$. Return happens for sure, but the **expected** wait is infinite, because rare very long excursions dominate the average. (This **null recurrence** is the same phenomenon that broke optional stopping in Chapter 3.)

Common pitfall “Returns with probability 1” and “returns quickly” are different statements. The symmetric walk satisfies the first ($\mathbb{P}(\tau_0 < \infty) = 1$) but flatly violates any finite-expectation reading: $\mathbb{E}[\tau_0] = \infty$. Probability-one events can have infinite expected waiting times.

4.5 Mini-lab: simulating gambler’s ruin

```
import numpy as np
rng = np.random.default_rng(3)

def ruin_trial(a, N, p):
    s, steps = a, 0
    while 0 < s < N:
        s += 1 if rng.random() < p else -1
        steps += 1
    return (s == N), steps # reached target?, duration

def estimate(a, N, p, trials=50_000):
    res = [ruin_trial(a, N, p) for _ in range(trials)]
    win = np.mean([r[0] for r in res])
    dur = np.mean([r[1] for r in res])
    return win, dur

# Symmetric: theory h = a/N = 0.5, D = a(N-a) = 2500
print(estimate(50, 100, 0.5)) # ~ (0.50, 2500)
# Biased against player p=0.49
qp = 0.51/0.49
h = ((qp**50) - 1)/((qp**100) - 1)
print("theory h =", h, " sim =", estimate(50, 100, 0.49)[0])
```

The empirical win rate and mean duration match the closed forms a/N and $a(N - a)$, and the biased case tracks the (q/p) formula — theory and simulation agree.

Exercises

16. A gambler starts with €30, bets €1 fair, and quits at €0 or €50. Find the probability of reaching €50 and the expected number of bets.
17. Repeat with a biased coin $p = 0.45$. Compute the ruin probability and compare with the fair case — by what factor does the bias change it?
18. Derive the symmetric gambler’s-ruin recurrence $h(a) = (h(a + 1) + h(a - 1))/2$ from first-step conditioning, and prove the only solution with $h(0) = 0, h(N) = 1$ is linear.

19. Using the martingale $M_n = (q/p)^{S_n}$ from Chapter 3 and optional stopping, re-derive the biased gambler's-ruin probability. (Apply $\mathbb{E}[M_\tau] = M_0$ at the absorbing time τ and solve for the hitting probability.)

20. Explain, in terms of $D(a) = a(N - a)$, why a fair game between equally rich players ($a = N/2$) lasts longest, while a lopsided start ends quickly.

5 Markov Chains

5.1 Memorylessness and the transition matrix

A random walk's next position depends only on where it is now, not on how it got there. **Markov chains** generalise this **memoryless** property to any finite state space.

Definition 12 (Markov chain) A sequence $(X_n)_{n \geq 0}$ taking values in a finite set $\mathcal{S} = \{1, \dots, m\}$ is a **(time-homogeneous) Markov chain** if

$$\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1}, \dots, X_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i) =: P_{ij},$$

independent of n and of the past before time n . The numbers P_{ij} form the **transition matrix** P , a **stochastic matrix**: $P_{ij} \geq 0$ and every row sums to 1, $\sum_j P_{ij} = 1$.

Worked example — a two-state weather chain. States: Sunny (1), Rainy (2). A sunny day is followed by sun with prob 0.8, rain 0.2; a rainy day by sun 0.4, rain 0.6:

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}.$$

Each row sums to 1. The entry $P_{12} = 0.2$ is the chance sun turns to rain tomorrow.

5.2 Multi-step transitions are matrix powers

This is where MAT11-3 pays off. The probability of going from i to j in **exactly n steps** is the (i, j) entry of P^n .

Proposition 13 (Chapman–Kolmogorov) $\mathbb{P}(X_n = j \mid X_0 = i) = (P^n)_{ij}$. If the initial distribution is the row vector $\pi^{(0)}$, then the distribution at time n is $\pi^{(n)} = \pi^{(0)} P^n$.

Proof. Induction on n . For $n = 1$ it is the definition. Assuming it for n , condition on the state at time n (one step of partition refinement) and sum:

$$\mathbb{P}(X_{n+1} = j \mid X_0 = i) = \sum_k \mathbb{P}(X_n = k \mid X_0 = i) P_{kj} = \sum_k (P^n)_{ik} P_{kj} = (P^{n+1})_{ij},$$

which is matrix multiplication. $\square$

Worked example — two days ahead. For the weather chain,

$$P^2 = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}^2 = \begin{pmatrix} 0.72 & 0.28 \\ 0.56 & 0.44 \end{pmatrix}.$$

Starting sunny, the chance of sun two days later is $(P^2)_{11} = 0.72$.

5.3 Classifying states

The long-run behaviour of a chain is dictated by how its states connect.

Definition 14 (communication and classes) State j is **accessible** from i (written $i \rightarrow j$) if $(P^n)_{ij} > 0$ for some $n \geq 0$. States i and j **communicate** ($i \leftrightarrow j$) if each is accessible from the other. Communication is an **equivalence relation** (reflexive, symmetric, transitive — the structure from MAT11-4); its classes are the **communicating classes**. A chain is **irreducible** if there is a single class — every state reaches every other.

Definition 15 (recurrent, transient, absorbing) A state i is **recurrent** if, started from i , the chain returns to i with probability 1; otherwise it is **transient** (positive probability of never returning). A state i is **absorbing** if $P_{ii} = 1$ — once entered, never left. Absorbing states are the extreme case of recurrence.

Definition 16 (period and aperiodicity) The **period** of state i is $d(i) = \gcd\{n \geq 1 : (P^n)_{ii} > 0\}$. If $d(i) = 1$ the state is **aperiodic**. States in one communicating class share a period; an irreducible chain is **aperiodic** if that common period is 1.

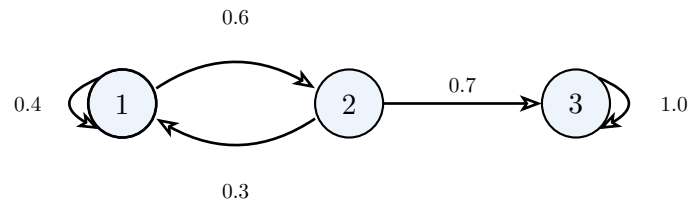


Figure 4: A chain with a transient class $\{1, 2\}$ leaking into an absorbing state 3. From 1 or 2 the chain eventually falls into 3 and stays; 3 is recurrent (indeed absorbing), 1 and 2 are transient.

Worked example — reading the diagram. For the chain above, $P = \begin{pmatrix} 0.4 & 0.6 & 0 \\ 0.3 & 0 & 0.7 \\ 0 & 0 & 1 \end{pmatrix}$. State 3 is absorbing ($P_{33} = 1$). From 1 and 2 the chain can reach 3 but never comes back, so 1, 2 are **transient**; 3 is **recurrent**. The chain is **not irreducible** (two classes: $\{1, 2\}$ and $\{3\}$). Eventually the chain is absorbed at 3 with probability 1.

Common pitfall “Recurrent” is about **returning**, not about **staying**. An absorbing state is recurrent, but a non-absorbing recurrent state is left and re-entered infinitely often. Conversely, a state you visit **many times early on** can still be transient if there is **any** escape route with positive probability — transience is decided by the long run, not the short run.

5.4 Stationary distributions

A **stationary distribution** is a probability row vector that the chain leaves unchanged: the fractions of time spent in each state, once the chain has “forgotten” its start.

Definition 17 (stationary distribution) A probability vector $\pi = (\pi_1, \dots, \pi_m)$ (entries ≥ 0 , summing to 1) is **stationary** for P if

$$\pi P = \pi, \quad \text{i.e.} \quad \pi_j = \sum_i \pi_i P_{ij} \text{ for all } j.$$

It is a **left eigenvector of P with eigenvalue 1** — an idea straight from MAT11-3.

Worked example — long-run weather. Solve $\pi \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix} = \pi$ with $\pi_1 + \pi_2 = 1$. The first equation gives $0.8\pi_1 + 0.4\pi_2 = \pi_1$, i.e. $0.4\pi_2 = 0.2\pi_1$, so $\pi_1 = 2\pi_2$. With $\pi_1 + \pi_2 = 1$ we get

$$\pi = \left(\frac{2}{3}, \frac{1}{3}\right).$$

In the long run it is sunny two-thirds of the days, regardless of today's weather.

The fundamental convergence theorem ties everything together.

Theorem 18 (convergence to stationarity) If a finite Markov chain is **irreducible** and **aperiodic**, then it has a **unique** stationary distribution π , and from **any** starting distribution

$$\pi^{(n)} = \pi^{(0)} P^n \rightarrow \pi \quad \text{as } n \rightarrow \infty.$$

Moreover $\pi_i = 1/\mathbb{E}[T_i]$, where T_i is the return time to state i : the long-run fraction of time in i is the reciprocal of its mean return time.

We state this without the full proof (it belongs to the spectral theory of non-negative matrices); the two-state example already shows P^n converging to the matrix with both rows equal to π .

Common pitfall Both hypotheses are needed. A chain with two absorbing states is **reducible** and has **infinitely many** stationary distributions (any mix of the two). A periodic chain — e.g. $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which just alternates — **has** a unique stationary distribution $(1/2, 1/2)$ but $\pi^{(n)}$ **oscillates** and never converges. “Stationary distribution exists” and “the chain converges to it” are distinct claims; aperiodicity is what upgrades the first to the second.

5.5 Mini-lab: stationary distribution by simulation and by eigenvector

```
import numpy as np

P = np.array([[0.8, 0.2],
              [0.4, 0.6]])

# (1) Power iteration: pi^(0) P^n
pi = np.array([1.0, 0.0])          # start certain-sunny
for _ in range(50):
    pi = pi @ P
print("power iteration :", pi)      # ~ [0.6667 0.3333]

# (2) Left eigenvector with eigenvalue 1
vals, vecs = np.linalg.eig(P.T)
v = np.real(vecs[:, np.argmin(np.abs(vals - 1))])
print("eigenvector      :", v / v.sum()) # ~ [0.6667 0.3333]
```

```
# (3) Empirical occupation frequency
rng = np.random.default_rng(4)
state, counts = 0, np.zeros(2)
for _ in range(1_000_000):
    counts[state] += 1
    state = rng.choice(2, p=P[state])
print("occupation freq :", counts / counts.sum()) # ~ [0.6667 0.3333]
```

All three routes — matrix powers, the eigenvector $\pi P = \pi$, and the empirical fraction of time spent in each state — agree on $\pi = (2/3, 1/3)$, the three faces of the convergence theorem.

Exercises

21. A frog hops among three lily pads with $P = \begin{pmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{pmatrix}$. Find the period of each state.

Is the chain irreducible? Aperiodic? Does $\pi^{(n)}$ converge?

22. Compute P^2 and P^3 for the weather chain and watch the rows approach $(2/3, 1/3)$. How many steps until both rows agree to two decimals?

23. For the absorbing chain $P = \begin{pmatrix} 0.4 & 0.6 & 0 \\ 0.3 & 0 & 0.7 \\ 0 & 0 & 1 \end{pmatrix}$, set up and solve the linear system for the absorption probability into state 3 from each transient state. (Here it is 1 from both — verify, then modify P to add a second absorbing state and redo.)

24. Prove that communication ($i \leftrightarrow j$) is an equivalence relation, using accessibility through intermediate states (Chapman–Kolmogorov) for transitivity.

25. Find all stationary distributions of $P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and of $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. For each, state whether $\pi^{(n)}$ converges and connect your answer to irreducibility and aperiodicity.

6 Concentration Inequalities

6.1 The question

Knowing only the mean — or the mean and variance — of a random variable, how confident can we be that it lands **near** its mean? **Concentration inequalities** answer this with universal bounds, requiring no knowledge of the full distribution. They are the quantitative engine behind every “with high probability” statement in data science.

6.2 Markov’s inequality

The weakest assumption — **non-negativity and a known mean** — already forbids too much mass far out in the tail.

Theorem 19 (Markov’s inequality) If $X \geq 0$ and $a > 0$, then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Proof. Compare X with the indicator that overshoots a . For every outcome,

$$a \cdot \mathbb{1}\{X \geq a\} \leq X,$$

because when $X \geq a$ the left side is $a \leq X$, and when $X < a$ the left side is $0 \leq X$. Take expectations (monotonicity, then linearity):

$$a \mathbb{P}(X \geq a) = a \mathbb{E}[\mathbb{1}\{X \geq a\}] \leq \mathbb{E}[X],$$

and divide by $a > 0$. $\square$

Worked example — income tail. If the average income in a population is €40{,}000, then at most $40000/200000 = 20\%$ of people can earn at least €200{,}000 — five times the mean. No distributional assumption is needed; non-negativity and the mean suffice.

Common pitfall Markov’s inequality needs $X \geq 0$. Applied to a variable that can be negative it is false: with X symmetric about 0, $\mathbb{P}(X \geq a)$ can exceed $\mathbb{E}[X]/a = 0$. When X can be negative, apply Markov to $|X|$ or to $(X - \mathbb{E}[X])^2$ — which is exactly how Chebyshev’s inequality arises.

6.3 Chebyshev’s inequality

Add knowledge of the **variance** and you control deviations on **both** sides of the mean. Chebyshev is just Markov applied to the squared deviation.

Theorem 20 (Chebyshev’s inequality) For any X with finite mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \text{Var}(X)$, and any $k > 0$,

$$\mathbb{P}(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}.$$

Equivalently, with $k = t\sigma$, $\mathbb{P}(|X - \mu| \geq t\sigma) \leq 1/t^2$.

Proof. The event $|X - \mu| \geq k$ is identical to $(X - \mu)^2 \geq k^2$. The variable $(X - \mu)^2$ is non-negative, so Markov’s inequality applies with threshold k^2 :

$$\mathbb{P}((X - \mu)^2 \geq k^2) \leq \frac{\mathbb{E}[(X - \mu)^2]}{k^2} = \frac{\sigma^2}{k^2}. \square$$

$\square$

Worked example — the 2-sigma and 3-sigma bands. For **any** distribution with finite variance, at least $1 - 1/4 = 75\%$ of the mass lies within 2σ of the mean, and at least $1 - 1/9 \approx 89\%$ within 3σ . These hold with **no** normality assumption — far weaker than the “95% within 2σ ” of a normal, but universal.

Common pitfall Chebyshev is deliberately **loose**: it is the best bound using **only** the variance, so for any specific nice distribution it vastly overstates the tail. For a normal, $\mathbb{P}(|X - \mu| \geq 3\sigma) \approx 0.0027$, while Chebyshev only promises $\leq 1/9 \approx 0.11$ — a factor of 40. Use Chebyshev when you know **nothing** but the variance; expect much sharper bounds (next) once independence and boundedness are available.

6.4 Hoeffding’s inequality

For a **sum (or average) of independent bounded variables**, the tail decays not like $1/k^2$ but **exponentially**. This is the qualitative leap that makes large samples so powerful.

Theorem 21 (Hoeffding's inequality, stated) Let $X_1, \dots, X_n$ be independent with $a_i \leq X_i \leq b_i$, and let $S_n = \sum_{i=1}^n X_i$. Then for any $t > 0$,

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

For the **average** $\bar{X} = S_n/n$ of variables in $[a, b]$,

$$\mathbb{P}(|\bar{X} - \mathbb{E}[\bar{X}]| \geq \varepsilon) \leq 2 \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right).$$

We **state** Hoeffding's bound and **verify it empirically** (below) rather than prove it; the proof uses the exponential-moment (Chernoff) method, sketched next. The key features to absorb:

- the tail decays like $e^{-cn\varepsilon^2}$ — **exponential** in both the sample size n and the squared tolerance ε^2 ;
- it needs only **boundedness** and **independence**, not the shape of the distribution.

Remark Chernoff's method (the engine behind Hoeffding) bounds a tail by optimising over an exponential transform: for any $s > 0$,

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \geq t) \leq e^{-st} \mathbb{E}[e^{s(S_n - \mathbb{E}[S_n])}] = e^{-st} \prod_i \mathbb{E}[e^{s(X_i - \mathbb{E}[X_i])}],$$

using Markov on e^{sS} and independence to factor. Bounding each factor (Hoeffding's lemma: a variable in $[a, b]$ has $\mathbb{E}[e^{s(X - \mathbb{E}X)}] \leq e^{s^2(b-a)^2/8}$) and minimising over s yields the exponential bound. The single trick — **apply Markov to e^{sX} and optimise s** — is worth remembering; it converts a polynomial Chebyshev tail into an exponential one.

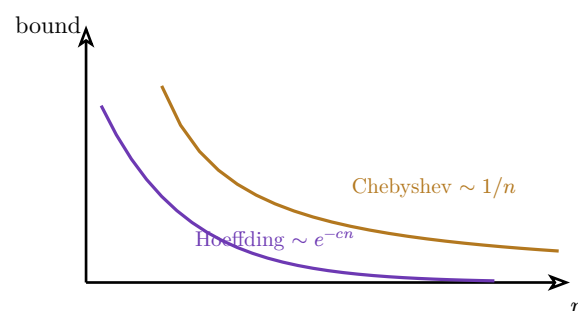


Figure 5: Tail probability $\mathbb{P}(\bar{X} - \mu \geq \varepsilon)$ for an average of n bounded variables: Chebyshev decays like $1/(n\varepsilon^2)$ (slow), Hoeffding like $e^{-2n\varepsilon^2}$ (fast). The true tail sits below both.

6.5 Mini-lab: how tight are the bounds?

```
import numpy as np
rng = np.random.default_rng(5)

# Average of n fair coin flips in [0,1]; mu = 0.5
def tail_estimate(n, eps, trials=400_000):
    means = rng.binomial(n, 0.5, trials) / n
    emp = np.mean(np.abs(means - 0.5) >= eps)
    cheb = min(1.0, 0.25 / (n * eps**2))
    hoef = 2 * np.exp(-2 * n * eps**2)
    # var of one flip = 1/4
    # (b-a)=1
```

```

    return emp, cheb, hoef

for n in (50, 200, 1000):
    emp, cheb, hoef = tail_estimate(n, 0.1)
    print(f"n={n:5d}:  empirical={emp:.5f}  Hoeffding={hoef:.5f}
    Chebyshev={cheb:.5f}")

```

You will see `empirical <= Hoeffding <= Chebyshev` at every n : both bounds are valid (never violated), Hoeffding is far tighter, and as n grows the empirical tail and the Hoeffding bound both collapse exponentially while Chebyshev limps down like $1/n$.

Exercises

26. A non-negative variable has mean 5. Bound $\mathbb{P}(X \geq 20)$ by Markov. If you also learn $\text{Var}(X) = 4$, bound $\mathbb{P}(|X - 5| \geq 15)$ by Chebyshev and compare.
27. Derive Chebyshev's inequality from Markov's in full detail, stating exactly where non-negativity is used.
28. For $n = 100$ independent variables in $[0, 1]$ with mean 0.5, use Hoeffding to bound $\mathbb{P}(\bar{X} \geq 0.6)$. Then bound the same event with Chebyshev (variance of each $\leq 1/4$) and report the ratio of the two bounds.
29. Show that Chebyshev is **tight**: construct a three-point distribution achieving $\mathbb{P}(|X - \mu| \geq k) = \sigma^2/k^2$ exactly. (Place mass at μ and at $\mu \pm k$.)
30. Empirically (or by hand for small n) confirm that Hoeffding is never violated for sums of ± 1 steps, and identify a regime where it is loose by a large factor compared with the true binomial tail.

7 From Bounds to Sample Sizes

7.1 Reading a concentration bound backwards

A concentration inequality, read forwards, says “**given** n , the error exceeds ε with at most probability δ .” Read **backwards**, it answers the question every experimenter actually asks: “**how large must n be** to guarantee error $\leq \varepsilon$ with confidence $1 - \delta$?” This inversion is the practical payoff of the whole course.

7.2 The Hoeffding sample-size formula

Proposition 22 (sample size from Hoeffding) Let $\bar{X}$ be the average of n independent variables in $[a, b]$ with mean μ . To guarantee

$$\mathbb{P}(|\bar{X} - \mu| \geq \varepsilon) \leq \delta,$$

it suffices to take

$$n \geq \frac{(b - a)^2 \ln(2/\delta)}{2\varepsilon^2}.$$

Proof. Hoeffding gives $\mathbb{P}(|\bar{X} - \mu| \geq \varepsilon) \leq 2 \exp(-2n\varepsilon^2/(b - a)^2)$. Set the right side $\leq \delta$ and solve: $2e^{-2n\varepsilon^2/(b-a)^2} \leq \delta \Leftrightarrow e^{-2n\varepsilon^2/(b-a)^2} \leq \delta/2 \Leftrightarrow 2n\varepsilon^2/(b - a)^2 \geq \ln(2/\delta)$, giving the stated n . $\square$ $\square$

Notice the **scaling laws** baked into the formula, worth committing to memory:

- halving the tolerance ε quadruples n (the $1/\varepsilon^2$): precision is expensive;

- **raising confidence costs only logarithmically** ($\ln(2/\delta)$): going from 95% to 99% barely moves n .

Worked example — polling. Estimate the fraction p voting “yes” (each response in $\{0, 1\}$, so $b - a = 1$) to within $\varepsilon = 0.03$ with 95% confidence ($\delta = 0.05$):

$$n \geq \frac{1 \cdot \ln(2/0.05)}{2 \cdot 0.03^2} = \frac{\ln 40}{0.0018} = \frac{3.689}{0.0018} \approx 2050.$$

About 2000 respondents suffice — **regardless of the population size** and with no assumption beyond independence and boundedness. This is why national polls use samples of one to two thousand.

Worked example — the price of precision. Tighten the poll to $\varepsilon = 0.01$ (one percentage point), same confidence:

$$n \geq \frac{\ln 40}{2 \cdot 0.01^2} = \frac{3.689}{0.0002} \approx 18450.$$

Nine times the tolerance reduction ($3 \times$ smaller ε) costs $9 \times$ the sample — the $1/\varepsilon^2$ law in action.

Common pitfall The Hoeffding sample size is a **worst-case** (distribution-free) guarantee, so it is **conservative**: it assumes the most adversarial distribution on $[a, b]$. If you genuinely know more — a small variance, normality — sharper formulas (Chebyshev with the true σ^2 , or a normal-approximation z -interval) give smaller n . Never report the Hoeffding n as “the” required sample size without noting it is an upper bound that buys **guaranteed** coverage at the cost of extra observations.

7.3 Chebyshev sample sizes, for comparison

If you know the variance σ^2 but not boundedness, invert Chebyshev instead: $\mathbb{P}(|\bar{X} - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2) \leq \delta$ forces

$$n \geq \frac{\sigma^2}{\delta \varepsilon^2}.$$

Here n scales like $1/\delta$ — **linearly** in the confidence parameter, far worse than Hoeffding’s $\ln(1/\delta)$. For high confidence (small δ) the exponential bound wins enormously; this is precisely why Hoeffding/Chernoff bounds dominate modern statistics and machine learning.

Worked example — Chebyshev vs Hoeffding on the same poll. For the poll ($\varepsilon = 0.03$, $\delta = 0.05$), Chebyshev with the worst-case Bernoulli variance $\sigma^2 \leq 1/4$ requires

$$n \geq \frac{1/4}{0.05 \cdot 0.03^2} = \frac{0.25}{0.000045} \approx 5556,$$

versus Hoeffding’s ≈ 2050 . Same guarantee, less than half the data — the exponential bound’s $\ln(1/\delta)$ beats Chebyshev’s $1/\delta$.

7.4 Mini-lab: does the prescribed sample size deliver?

```
import numpy as np
rng = np.random.default_rng(6)
```

```

def required_n(eps, delta, span=1.0):
    return int(np.ceil(span**2 * np.log(2/delta) / (2*eps**2)))

eps, delta, p = 0.03, 0.05, 0.5
n = required_n(eps, delta)
print("prescribed n =", n)                # ~ 2050

# Run many polls of size n; check the failure rate stays <= delta
trials = 20_000
phat = rng.binomial(n, p, trials) / n
fail = np.mean(np.abs(phat - p) >= eps)
print(f"observed failure rate = {fail:.4f} (target <= {delta})")

```

The observed failure rate comes out **well under** $\delta = 0.05$ — typically a fraction of a percent — confirming both that the formula works and that it is conservative: Hoeffding hands you a safety margin you pay for in extra samples.

Exercises

- 31.** How many independent bounded $([0, 1])$ samples guarantee the sample mean is within $\varepsilon = 0.02$ of the true mean with 99% confidence? With 95%? By what factor does the stricter confidence raise n — and is the factor large or small? Explain via the $\ln(1/\delta)$ scaling.
- 32.** A sensor reading lies in $[0, 10]$. How many readings are needed to estimate the mean within $\varepsilon = 0.1$ units at 95% confidence? Note how the span $(b - a) = 10$ enters.
- 33.** Show that to **halve** the error tolerance you must **quadruple** the sample, but to **square** the confidence ($\delta \rightarrow \delta^2$) you only **double** the $\ln(2/\delta)$ -term approximately. Quantify both from the formula.
- 34.** For estimating a proportion with $\sigma^2 \leq 1/4$, tabulate the Chebyshev and Hoeffding sample sizes at $\delta \in \{0.1, 0.05, 0.01, 0.001\}$ and $\varepsilon = 0.03$. Plot n versus δ for both and explain the diverging shapes ($1/\delta$ vs $\ln(1/\delta)$).
- 35.** You ran $n = 1000$ bounded- $[0, 1]$ samples and got $\bar{X} = 0.62$. Using Hoeffding, state a 95% confidence interval for μ (invert the bound for ε at fixed n, δ). How wide is it?

8 Epilogue: where this leads

You now hold a coherent toolkit for **discrete stochastic reasoning**. Conditional expectation turned conditioning into an algebraic object obeying the tower property; the law of total variance split uncertainty into explained and residual parts; martingales formalised the fair game and its no-free-lunch optional-stopping behaviour; random walks and Markov chains gave concrete dynamic models with hitting times, absorption, and stationary distributions; and concentration inequalities converted means and variances into quantitative, invertible guarantees on uncertainty.

Three doors open ahead. The **information-theoretic reasoning** of S3 will recast “conditioning as information gain” in the precise currency of entropy and mutual information. The **stochastic calculus** of S6 will rebuild martingales on the formal foundation of sigma-algebras and filtrations — the apparatus we deliberately set aside here — and extend random walks to Brownian motion. And every applied course that says “**with high probability**” leans on exactly the Markov–Chebyshev–Hoeffding chain you derived in Chapter 6. The partitions you started with in MAT11-4 have grown into a working theory of uncertainty.